6. The set-up in Figure 6.1 is to find the speed of sound in air. Two identical microphones A and B are connected to a timer and placed on a bench top as shown. The timer can be triggered to 'start' and 'stop' timing using the respective microphones to feed signals to the START and STOP terminals of the timer.

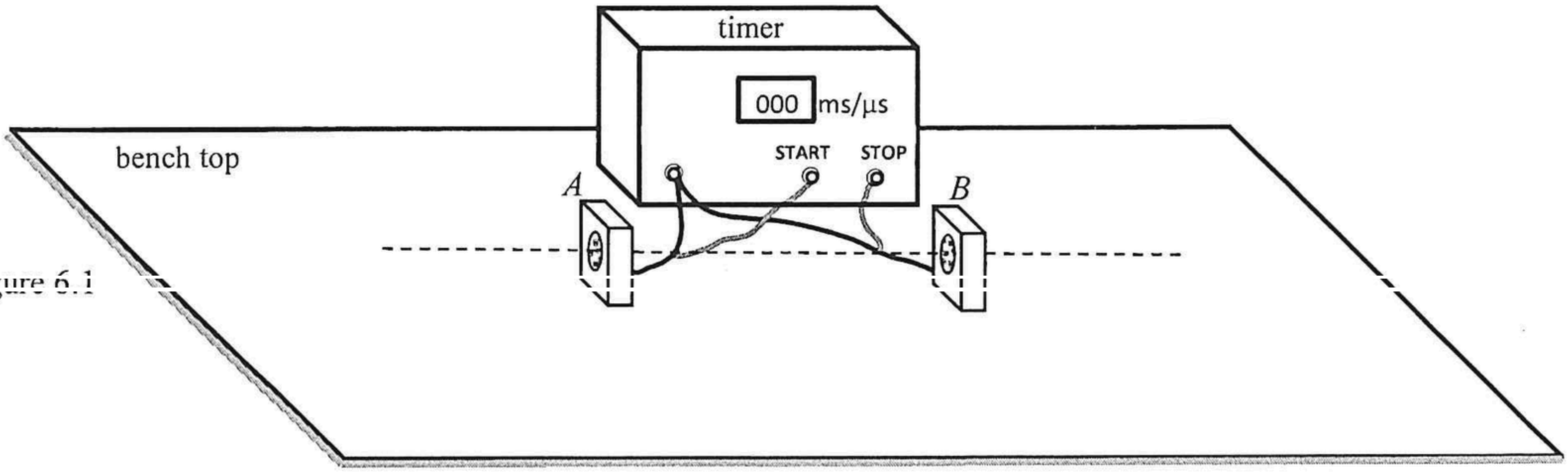
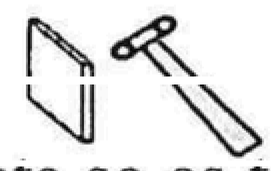


Figure 6.1

(a) You are given a hammer and a metal plate (). Use 'X' to indicate a suitable location on Figure 6.1 where the hammer should hit the plate so as to generate a sharp loud sound to be received by the microphones in this experiment. State an additional piece of apparatus needed and the measurements to be made in this experiment. (3 marks)

(b) The separation between A and B is set at 0.280 m. The experiment is repeated to obtain a few readings of the timer as follows:

801 μs, 838 μs, 539 μs, 821 μs

(i) Find the speed of sound in air. Show how you would treat the data obtained in the calculation.

(ii) Suggest one adjustment to the experimental setting so as to obtain a more accurate result.

(3 marks)

Answers written in the margins will not be marked.

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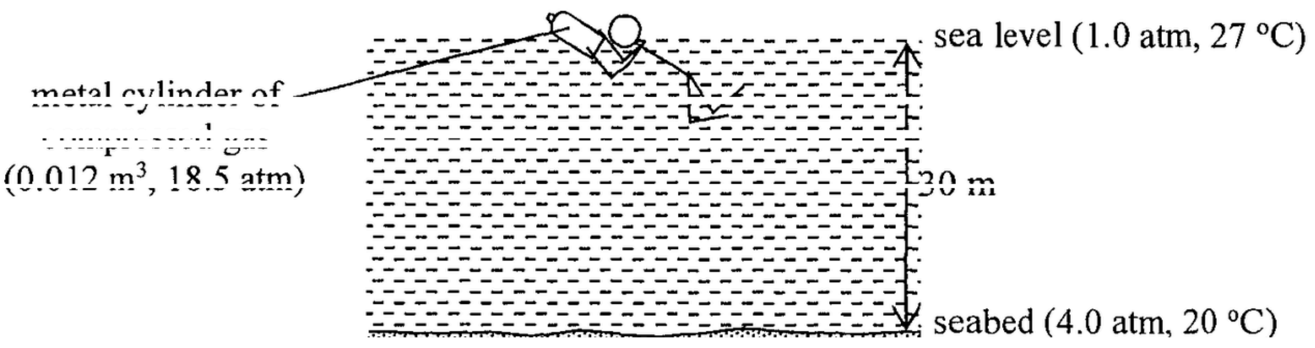
2. A diver makes a sound by tapping a metal cylinder at sea level. Within a time of 0.04 s, the sound goes vertically to the seabed 30 m below and echoes back to the sea level.

(a) Estimate the speed of sound in sea water. (2 marks)

Figure 2.1



The metal cylinder of volume $0.012\ m^3$ contains compressed gas under a pressure of 18.5 atm is initially at sea level, where the pressure is 1.0 atm and the temperature is 27 °C. The diver then brings the cylinder to the seabed where the pressure is 4.0 atm and the temperature is 20 °C. Assume that the volume of the cylinder remains unchanged. Given: atmospheric pressure 1.0 atm = 1.01×10^5 Pa

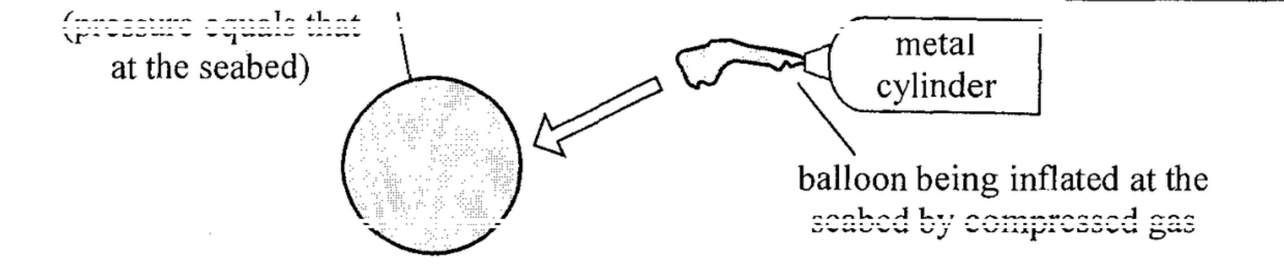
*(b)(i) Show that at the seabed the pressure in the cylinder becomes 18.1 atm. (1 mark)

(ii) Explain the pressure drop in the cylinder using the kinetic theory. (2 marks)

*(c) The diver then inflates identical balloons each to a volume of $0.015\ m^3$ by using the cylinder of compressed gas at the seabed. Assume that the balloons are inflated slowly so that the temperature of the gas remains unchanged and the final pressure in the balloon equals that at the seabed.

NOT drawn to scale

Figure 2.2

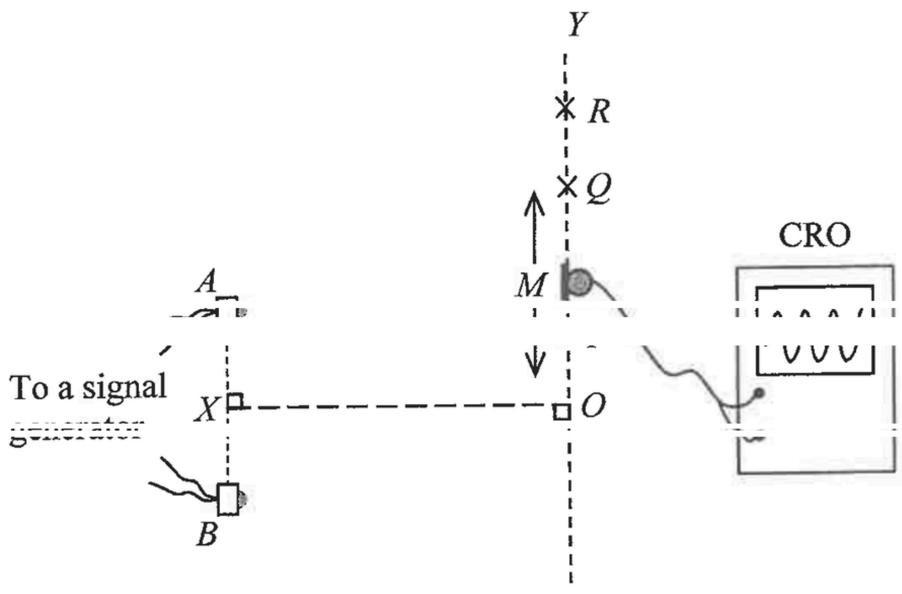


(i) Show that the gas pressure in the cylinder decreases by 5.0 atm after inflating one balloon. (2 marks)

(ii) Hence, find the total number of balloons that the diver can inflate completely. (2 marks)

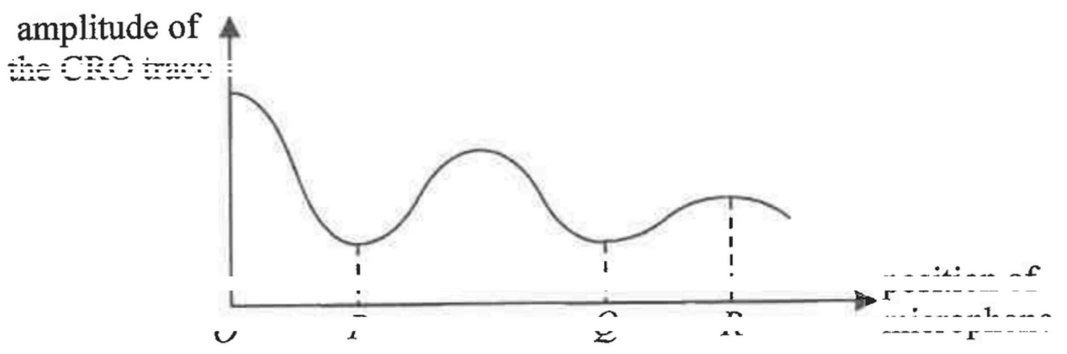
6.

Figure 6.1



In Figure 6.1, two small identical loudspeakers A and B produce coherent sound waves. X is the mid-point of AB. A microphone M connected to a CRO is moved along OY to detect the loudness of the sound, with CRO trace of a larger amplitude representing a greater loudness. Figure 6.2 shows the result.

Figure 6.2



(a) Explain what is meant by **coherent** sound waves. (1 mark)

(b) (i) Explain why sound of alternate maximum and minimum loudness is detected along OY. (2 marks)

(ii) The amplitude of the CRO trace at P is **not** zero. Suggest a possible reason. (1 mark)

(c) Given: $AQ = 2.17$ m, $BQ = 2.58$ m
Find the speed of sound in air if the frequency of the signal generator is 1200 Hz. (2 marks)

(d) Given that the separation of A and B is 0.80 m, explain why no more maximum can be detected when the microphone is moved beyond position R along OY. (2 marks)

(e) The microphone is now moved from O to X along the line OX. State whether the amplitude of the CRO trace increases, decreases, remains the same, or varies periodically. (1 mark)

5. A sound wave from a loudspeaker is propagating to the right in air. Figure 5.1(a) shows the equilibrium positions of some particles (a to u) in air. Figure 5.1(b) shows their positions at time $t = 0$.

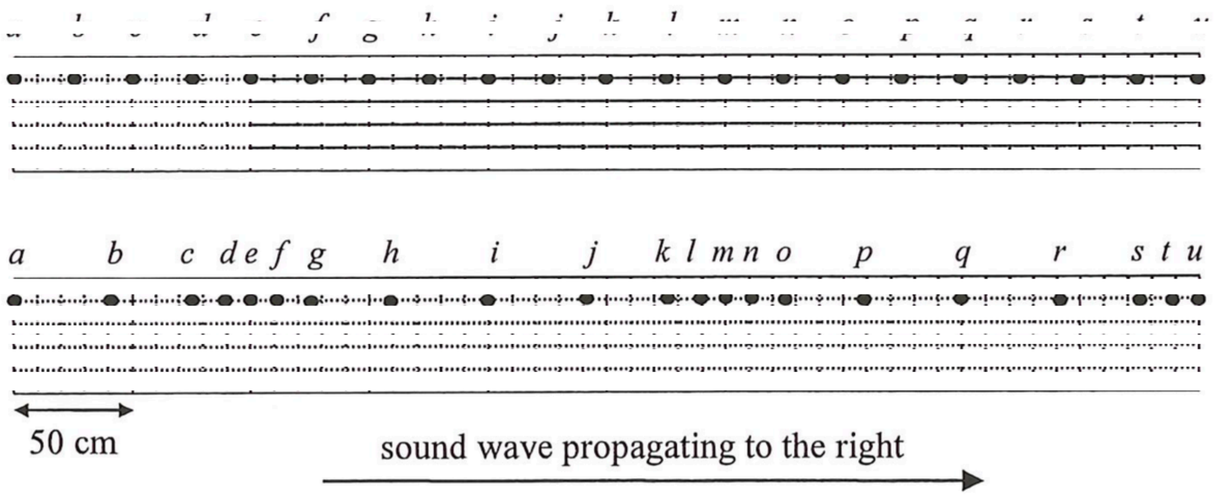


Figure 5.1(a)

$t = 0$ Figure 5.1(b)

(a) At time $t = 0$, state a particle which

(i) is at a centre of rarefaction. (1 mark)

(ii) has the greatest displacement to the right from its equilibrium position. (1 mark)

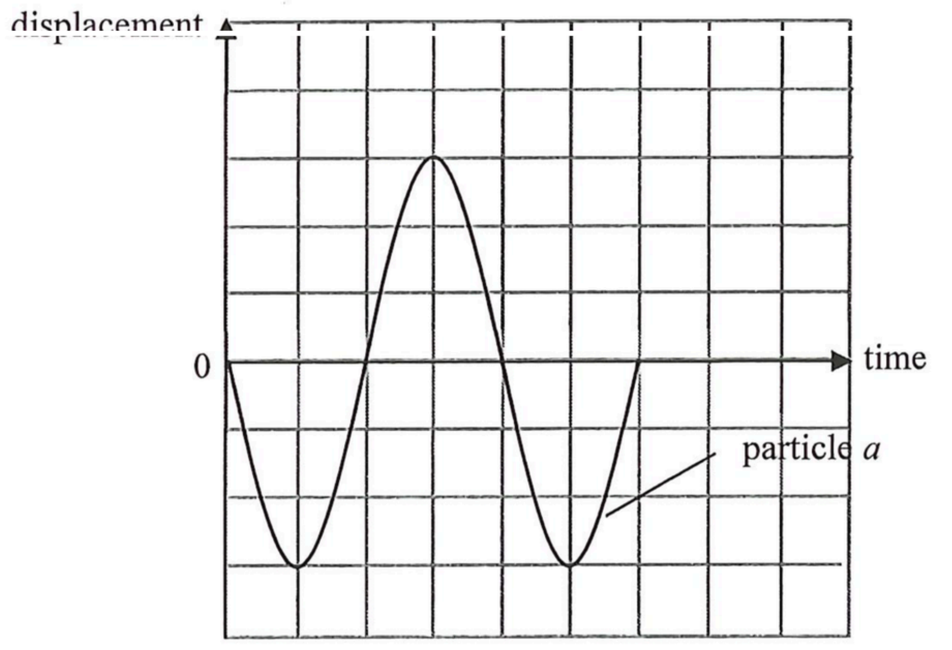
(b) The width of each grid is 50 cm as shown in Figure 5.1(b). The speed of sound in air is 340 m s^{-1}.

(i) What is the wavelength of the sound ? (1 mark)

(ii) Find the frequency of the sound. (2 marks)

(iii) Figure 5.2 shows the displacement-time graph of particle a. Sketch on the same figure the corresponding graph for particle m. (2 marks)

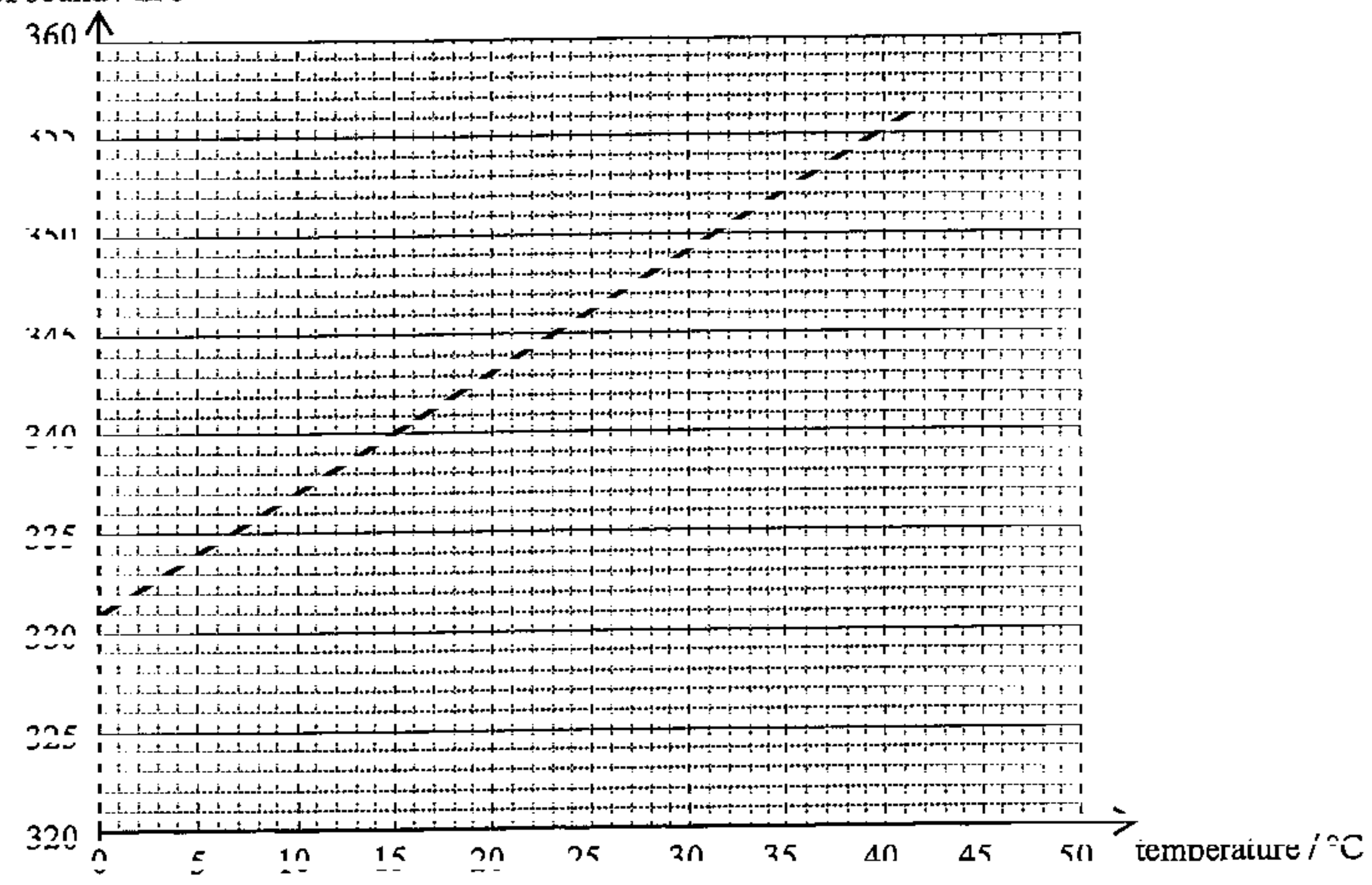
Figure 5.2



(c) State the change on the position of particle h at time $t = 0$ if the loudspeaker is turned louder. (1 mark)

7. The graph below shows the speed of sound at different air temperatures.

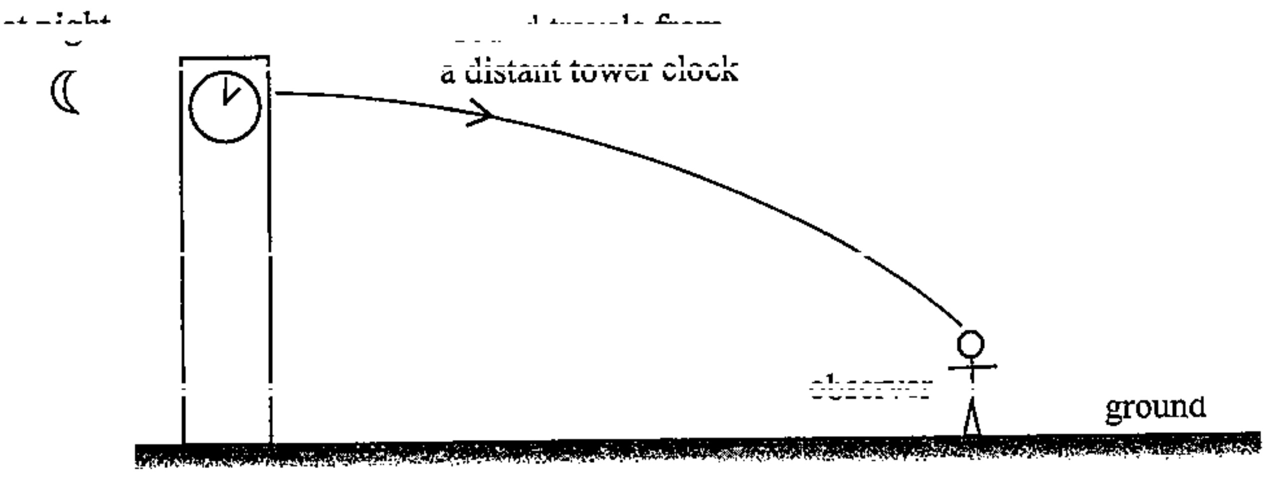
Figure 7.1



(a) With reference to the graph, state the relationship between the speed of sound and the air temperature. (1 mark)

(b) Figure 7.2 shows sound travelling from a distant tower clock to an observer at night when the ground cools.

Figure 7.2



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How does the temperature difference between air layers above the ground explain the situation in Figure 7.2 ? (2 marks)

(c) It is known that the air temperature is 10 °C. Referring to the graph in Figure 7.1:

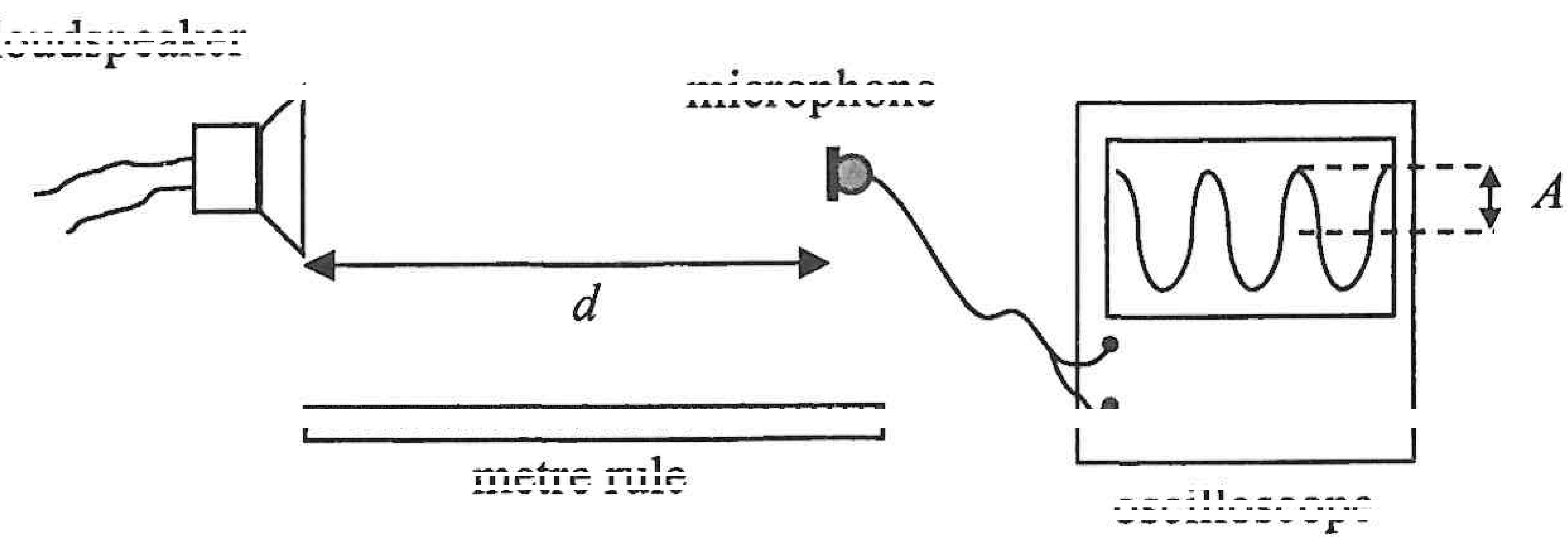
(i) estimate the ratio of speed of light to the speed of sound; (1 mark)

(ii) estimate the distance of a thunderstorm from an observer O if thunder is heard 3.0 s after lightning is seen by O. (2 marks)

Answers written in the margins will not be marked.

6. A student uses the set-up in the figure below to investigate the relation between the loudness of sound and the distance from the source of sound. The distance between the loudspeaker and the microphone is d. The loudness of the sound relates to the amplitude A of the oscilloscope trace.

The student hypothesises that A is inversely proportional to d^2 ($A \propto \frac{1}{d^2}$).



Describe

(I) the procedure of the experiment;

(II) how the data should be analysed using a graphical method to verify whether the hypothesis ($A \propto \frac{1}{d^2}$) is valid.

(5 marks)

Answers written in the margins will not be marked.